Large-Neighborhood Search as Audited Plan Improvement

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Abstract

This draft documents the BISECT local-search and large-neighborhood-search improvement contract. It treats `bisect improve` as a lineage-preserving heuristic implementation: improved plans, rejected moves, no-op outcomes, and staged LNS/tabu paths must all produce reproducible summaries. Accepted improvements must produce RPLAN/RCTX descendants that name the parent plan, child artifact, method, status, objective delta, and validation profile.

1 Introduction

U.18 documents `bisect-local-search` and `bisect-cli::improve_cmd`, exposed as `bisect improve` with staged `--search lns` and `--search tabu` surfaces. The paper is a heuristic improvement implementation paper, not a global optimality claim.

2 Implementation Contract

The current implementation centers on a one-move boundary improvement baseline plus structured statuses for no-improvement, rejected-move, and staged-method paths. The summary must record parent plan id, attempted method, accepted move or rejection/no-op reason, objective delta, validity checks, and output artifact id when a child plan is emitted.

The paper therefore does not treat every command invocation as a successful improvement. A no-op transcript is evidence that the search was run under a declared method and profile. A rejected-move transcript is evidence that the implementation preserved validity by refusing a candidate. Only an *Improved* status supports the claim that a descendant plan was emitted with a measured local objective delta.

3 Audit Artifacts

Improved plans are descendants of an existing RPLAN input. The RCTX and audit certificate should preserve parent/child lineage, declared profile checks, and the improvement

Outcome	Required summary	Claim supported
Improved	parent, child, move, delta, validity checks	local improvement
Rejected move	parent, candidate move, rejection reason	validity-preserving search
No-op	parent, search status, reason	reproducible non-improvement
Staged method	requested method and unsupported scope	roadmap boundary

Table 1: Improvement-family statuses.

summary so a verifier can inspect both validity and provenance. The audit package must identify the parent RPLAN, emitted child RPLAN when one exists, method, status, objective before and after, objective delta, validation profile, and any random seed used by the method.

Validity and improvement are separate claims. A valid child plan is not automatically a meaningful improvement, and an apparent objective improvement is not accepted unless the declared profile checks pass. No-op and staged-method records still belong in the audit trail because they explain why no descendant artifact was produced.

4 Evaluation Plan

Current evidence is L0 improvement/no-op/rejected-move/staged-method behavior and L1 CLI/RPLAN sidecar tests. Empirical distributions of accepted moves, objective deltas, and tabu/LNS comparisons are future evidence.

Claim	Current evidence	Publication gap
One-move search can emit valid descendants	L0 improvement fixtures and L1 RPLAN sidecar tests	Public before/after package
No-op and rejected moves are reproducible	L0 status fixtures	CLI transcripts for each status
Staged LNS/tabu surfaces are honest roadmap boundaries	Staged-method status tests	Full method implementation
Large-instance heuristic quality improves materially	Future evidence only	Benchmark protocol and comparison baselines

Table 2: Claim and evidence ladder for U.18.

5 Limitations

One-move local search can get stuck in local optima and should not be reported as global search. Advanced LNS and tabu methods are scaffolded until their performance protocols and validity-preserving repair hooks are documented. The current paper supports an audited improvement contract, not a claim that the implementation finds globally optimal or even consistently high-quality plans across large instances.

The future empirical paper must include method parameters, random seeds where relevant, acceptance rules, distributions of accepted moves and objective deltas, and comparisons against no-op, one-move, tabu, and LNS baselines before making broader performance claims.

6 Conclusion

U.18 makes plan improvement auditable: the output is not just a better score, but a child plan with a recorded parent, method, status, objective delta, and verifier path. The reviewed draft now treats no-op, rejected, and staged paths as evidence-bearing outputs while reserving LNS/tabu performance claims for a later benchmarked implementation.

References

BISECT project. U.18 large-neighborhood search paper plan, 2026. Local planning document and implementation roadmap.